

# BioSpecInfo

## Traceability Matrix — BioSpecInfo

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# Traceability Matrix — BioSpecInfo

| Field             | Value                                                                                                        |
|-------------------|--------------------------------------------------------------------------------------------------------------|
| Software          | BioSpecInfo                                                                                                  |
| Author            | Samuele Pio Provenzano                                                                                       |
| Version described | bsi-v175                                                                                                     |
| Purpose           | Connect every declared requirement to the implementation that realises it and to the bench that verifies it. |

**How to read this matrix.** Every row is a closed chain: a requirement, the point in the code that realises it, and the bench that verifies it works. A row without a bench is an **unverified** requirement and is declared as such, not hidden: the usefulness of a matrix lies precisely in making the gaps visible.

The outcomes of the benches cited here are reported in full, with their complete output, in [../evidence/RAPPORTO-VERIFICA.md](#), which is generated automatically.

## 1. Scientific requirements

| ID     | Requirement                                                                                  | Implementation                                                                                                   | Verification bench                                                                         |
|--------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| SCI-01 | Structural data for drugs must be consistent with the declared molecular weight              | <code>FARM_DATA</code> in <code>index.html</code>                                                                | <code>tools/verifica-farmaci.js</code> — RDKit<br>↔ literature comparison over 178 entries |
| SCI-02 | Functional groups must be recognised on the structure, not on the text of the SMILES         | <code>bsi-spettri.js</code> §1, SMARTS patterns via RDKit                                                        | <code>test_spettri</code> — 8 reference molecules, additive recognition                    |
| SCI-03 | A molecule with several functional groups must show them all                                 | <code>bandeDaGruppi()</code> , additive construction                                                             | <code>test_spettri</code> §1 — acetylsalicylic acid: ester <b>and</b> acid                 |
| SCI-04 | Spectra must respect the conventions of their technique                                      | <code>svgIR()</code> , <code>makeNMRSvg()</code> , <code>drawUVSpectrum()</code> , <code>drawMSSpectrum()</code> | <code>test_assi</code> (SVG), <code>test_assi_canvas</code> (canvas)                       |
| SCI-05 | A computed spectrum must be reproducible                                                     | every <code>Math.random()</code> removed from the traces                                                         | <code>test_assi</code> §3, <code>test_spettri</code> §8 — byte-for-byte comparison         |
| SCI-06 | The height of <sup>1</sup> H peaks must represent the integration, or not be asserted at all | <code>makeNMRSvg()</code> ; <code>_nmrSvg()</code> draws bands of uniform height                                 | <code>test_assi</code> §4                                                                  |

| ID     | Requirement                                                                                                | Implementation                                                                                                        | Verification bench                                                                                                                                                                                |
|--------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI-07 | Conjugation must act on the individual carbonyl                                                            | SMARTS <code>acidoConiug</code> , <code>estereConiug</code> , <code>chetoneConiug</code> , <code>aldeideConiug</code> | <code>test_spettri §1</code> — aspirin 1760/1690; acetophenone 1690                                                                                                                               |
| SCI-08 | An ambiguous physical constant must not be resolved arbitrarily                                            | <code>costante_fisica</code> in <code>bsi-ai-hub.js</code> — collects every candidate                                 | <code>test_costanti</code> — 45 checks, including the refusals for ambiguity                                                                                                                      |
| SCI-09 | The limits of the predictors must be declared to the user                                                  | chart headers; <code>docs/06 §3.5</code>                                                                              | <code>verifica_guida</code> — consistency between promises and code                                                                                                                               |
| SCI-10 | Molecular symmetry must determine the selection rules correctly                                            | <code>ssimm</code> section in <code>index.html</code>                                                                 | <code>test_simmetria</code> — 26 checks: normal modes, centre of inversion, mutual exclusion                                                                                                      |
| SCI-12 | A chromatogram must follow from the parameters, not be drawn by hand                                       | <code>scroma</code> section: $\sigma = t_{R}/\sqrt{N}$ , $R_s$ from distance and widths                               | <code>audit_stabilita</code> (opening and drawing), inspection of the values against the definitions                                                                                              |
| SCI-13 | A calibration line must declare the uncertainty of its coefficients and recognise the two ways it misleads | <code>staratura</code> section: ordinary least squares, F-test line $\leftrightarrow$ parabola, deleted residual      | Verified on the three example sets included: linear ( $F = 0.3$ ), non-linear ( $F = 186.6 > 7.71$ ), with an outlier (1 detected)                                                                |
| SCI-14 | A QSAR model must be evaluated on scaffolds never seen, not on analogues from the same set                 | <code>bsi-cheminfo.js §9</code> — <code>divisionePerScaffold()</code> alongside <code>divisioneCasuale()</code>       | <code>test_cheminfo</code> — the scaffold split leaves no scaffold shared between training and test                                                                                               |
| SCI-15 | A model score must be compared against chance, not presented on its own                                    | <code>bsi-cheminfo.js §9</code> — retraining on scrambled labels, eight repetitions                                   | <code>test_cheminfo</code> — verified <b>in both directions</b> : on a learnable signal $R^2$ 0.394 beats all eight twins (margin 0.287); on random labels $R^2$ -0.199 does <b>not</b> beat them |
| SCI-16 | Two spellings of the same molecule must not be counted twice, nor land on opposite sides of the split      | <code>bsi-cheminfo.js §1</code> — deduplication on the canonical SMILES after the largest fragment                    | <code>test_cheminfo</code> — <code>OC(=O)C</code> and <code>CC(=O)O</code> collapse into one entry; salts lose the counterion                                                                     |
| SCI-17 | Similarity between molecules must be computed, not estimated                                               | <code>bsi-cheminfo.js §3</code> — Tanimoto and Dice over Morgan, RDKit, pattern and MACCS fingerprints                | <code>test_cheminfo</code> — compared against hand-computable values, including the $0/0 = 1$ convention                                                                                          |
| SCI-18 | A validation on a single split, over a small set, is a noisy number: it must come with its dispersion      | <code>bsi-cheminfo.js §9-bis</code> — <code>validazioneIncrociata()</code> over <b>scaffold-grouped</b> folds         | <code>test_cheminfo</code> — the folds cover every molecule once, <b>no scaffold sits in two folds</b> , and the same seed gives the same folds                                                   |

| ID     | Requirement                                                                 | Implementation                                                                                    | Verification bench                                                                                                                                                                              |
|--------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI-19 | A result that changes on every run is verifiable by nobody                  | seeded mulberry32 generator for splits, scrambles and folds; the seed is written into the report  | test_cheminfo — two calls with the same seed give identical folds; the method report declares the seed                                                                                          |
| SCI-20 | Results must be able to leave the tool, and the method must leave with them | bsi-cheminfo.js §9-ter — esportaCsv() and rapportoMetodo() , both formed in memory in the browser | test_cheminfo — columns consistent with the header, a name containing a comma quoted, residual computed, empty cell for missing activity; the report carries seed, fingerprint, verdict, limits |

## 2. AI agent requirements

| ID    | Requirement                                                                 | Implementation                                        | Verification bench                                             |
|-------|-----------------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------|
| AI-01 | A model retired by the provider must not block the assistant                | modelloSuggeritoDaErrore() , bsi_modelli_ko blacklist | test_404 — 36 checks                                           |
| AI-02 | A provider unreachable from the browser must be remembered and demoted      | bsi_prov_ko , 24 h TTL, providerUtilizzabili()        | test_ko (35), browser_ko (37)                                  |
| AI-03 | A temporary overload must produce a retry, not an error                     | erroreTemporaneo() , TENTATIVI_TEMPORANEO             | test_503 — 40 checks                                           |
| AI-04 | An imposed wait longer than 10 s must switch provider if a ready one exists | ATTESA_TROPPO_LUNGA_MS                                | test_attesalunga — 14 checks                                   |
| AI-05 | The provider's token ceiling must be learned from the error, not hard-coded | tettoDaErrore() , MARGINE_TETTO                       | test_tetto — 32 checks                                         |
| AI-06 | Opaque reasoning signatures must be returned to the provider                | capture and replay in appendAgentTurn()               | test_firma — 31 checks                                         |
| AI-07 | Two overlapping submissions must not corrupt the history                    | _turnoInCorso in send()                               | test_doppioinvio — 7 checks                                    |
| AI-08 | The user must be able to measure which providers they can actually reach    | provaTuttiIFornitori()                                | browser_prova — 29 checks                                      |
| AI-09 | Paid providers must not be offered as free ones                             | separation by PROVIDERS[id].free                      | browser_prova §D2                                              |
| AI-10 | A documented route must exist to remove CORS and rate limits                | proxy/spectra-proxy.js + in-app guide                 | test_guidaproxy (12), browser_proxy (15), browser_proxyui (17) |

### 3. Stability requirements

| ID     | Requirement                                                         | Implementation                                                | Verification bench                                                        |
|--------|---------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------|
| STA-01 | A long session must accumulate neither DOM nodes nor timers         | section lifecycle management                                  | <code>audit_stabilita</code> §1 — 88 sections × 5 rounds                  |
| STA-02 | Exhausting <code>localStorage</code> must not make the app unusable | guarded writes; persistent warning in the File Manager        | <code>audit_quota</code> (10),<br><code>test_filemanager</code> (15)      |
| STA-03 | No rejected promise may remain unhandled                            | systematic <code>.catch()</code>                              | <code>audit_promesse</code> — 22 checks, healthy network and dead network |
| STA-04 | Corrupted saved data must not prevent startup                       | <code>loadJSON()</code> with fallback                         | <code>audit_stabilita</code> §4                                           |
| STA-05 | A slow network must not be treated as an absent one                 | <code>ATTESA_RETE_MS = 3500</code> in <code>sw.js</code>      | <code>test_sw</code> §3 and §3-bis — measured 3,507 ms                    |
| STA-06 | An update must not destroy work in progress                         | <code>BSI_SW_UPDATED</code> message, decision taken page-side | <code>test_sw</code> §4                                                   |
| STA-07 | Browsing molecules must not exhaust the WebGL contexts              | reuse of the 3D viewer at all 5 creation points               | <code>test_visore3d</code> — from 7 constructions down to 1               |
| STA-08 | An unavailable IndexedDB store must be declared, not fatal          | <code>dbStore()</code> over a promise; persistent warning     | <code>test_filemanager</code> §2                                          |

### 4. Interface and accessibility requirements

| ID    | Requirement                                                                       | Implementation                                  | Verification bench                                                                                                                                                   |
|-------|-----------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UI-01 | Every page must open without JavaScript errors                                    | —                                               | <code>audit_stabilita</code> §5 — 14 pages                                                                                                                           |
| UI-02 | Charts must be sharp on high-density screens                                      | <code>bsiNitido()</code> on the canvas contexts | <code>audit_grafici</code> — 40 canvases                                                                                                                             |
| UI-03 | The app must work at 390 px width                                                 | auto-fit grids, no fixed column                 | <code>audit_mobile</code> — the document's <code>scrollWidth</code> across all 88 sections at 390 px;<br><code>audit_stabilita</code> for errors in a phone viewport |
| UI-04 | The user must be able to know which version they are running and force the update | "Updates" entry in the ✨ panel                  | <code>test_aggiorna</code> — 9 checks                                                                                                                                |
| UI-05 | Data deletion must be selective and reversible in its choices                     | <code>bsiCancellaDati()</code> by group         | <code>browser_reset</code> — 24 checks                                                                                                                               |

| ID    | Requirement                                                               | Implementation                                | Verification bench                                                                                                                                                            |
|-------|---------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UI-08 | Surfaces that appear only after an action must also meet WCAG AA contrast | the six panels of the Cheminformatics section | <code>test_cheminfo</code> — the analysis is run, the panels opened in turn, <b>460 text elements measured, 0 defects</b> , with the WCAG formulas rewritten inside the bench |

## 5. Security and privacy requirements

| ID     | Requirement                                                      | Implementation                                                    | Verification bench                                                                                                                                                                                                                                                                            |
|--------|------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEC-01 | No API key may be present in the repository                      | keys only in <code>localStorage</code> or in the Worker's secrets | <code>tools/verifica-sicurezza.js</code> — 235 tracked files, 8 credential shapes                                                                                                                                                                                                             |
| SEC-02 | No personal data may leave the device without an explicit action | local-first architecture, telemetry disabled                      | <code>audit_rete</code> — <b>direct verification</b> : a canary value seeded into 71 stores of the user's data, the application used across 88 sections on 6 pages, and the URL, headers and body of every request inspected. Plus <code>verifica-sicurezza</code> on the two exit mechanisms |
| SEC-03 | Passwords must not appear in clear text in the source            | SHA-256 in <code>file_manager.html</code>                         | <code>tools/verifica-sicurezza.js</code>                                                                                                                                                                                                                                                      |
| SEC-04 | No conflict marker may reach publication                         | —                                                                 | <code>verifica_guida</code> §12 — 58 text files                                                                                                                                                                                                                                               |

**SEC-03 — declared limitation.** GitHub Pages serves static files: any page-side access control can be bypassed by whoever reads the source. The File Manager's protection is a deterrent, **not** a security control, and the documentation says so. Furthermore the password remained in clear text in the git history until its removal, and taking a secret out of the files does not take it out of the history: `git log -p` hands it to anyone. The only effective remedy was to change it, and that **has been done** at version `bsi-v175`. The old one remains in the history and no longer opens anything.

## 6. Requirements not yet verified automatically

Declared for completeness. They are covered by documentary inspection or by manual testing, and their automation is planned.

| ID      | Requirement                                                      | Current coverage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UI-06   | Full WCAG 2.1 AA conformance                                     | the mechanical part is automated across <b>13 pages and 88 sections</b> ( <code>tools/verifica-accessibilita.js</code> , 33,642 text elements: it was 19,751 before text over gradients entered the measurement). <b>Of the 1,069 contrast defects that emerged, none remain: 0 measured</b> over those same 88 sections, and the value is recorded as a baseline the bench defends. Unlabelled fields are <b>zero</b> : see the box in <code>docs/09</code> §4. What remains outside is text inside SVGs and text over a real background image, counted on every run (D-09) |
| PERF-01 | First-paint time on a low-end device                             | manual cross-device testing ( <code>docs/02</code> §4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| SCI-11  | Structures of 2 entries that are not single molecules (it was 6) | <b>not representable</b> : Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. The other four were closed by taking the structure from ChEMBL, see <code>docs/06</code> §2.4                                                                                                                                                                                                                                                                                                                                                             |
| PERF-02 | Code coverage of the verification benches                        | <b>partially measured</b> : 49.79 % of statements over the widest path ( <code>audit_copertura</code> ). Whole-battery coverage and branch coverage remain unmeasured; see §8                                                                                                                                                                                                                                                                                                                                                                                                |
| UI-07   | Behaviour on Firefox and WebKit                                  | <b>not verified</b> — the battery runs on Chromium only; see §8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## 7. Overall coverage

| Category                 | Requirements | Verified by a bench | Coverage     |
|--------------------------|--------------|---------------------|--------------|
| Scientific (SCI-01...20) | 19           | 19                  | 100 %        |
| AI agent (AI-01...10)    | 10           | 10                  | 100 %        |
| Stability (STA-01...08)  | 8            | 8                   | 100 %        |
| Interface (UI-01...08)   | 6            | 6                   | 100 %        |
| Security (SEC-01...04)   | 4            | 4                   | 100 %        |
| <b>Automated total</b>   | <b>47</b>    | <b>47</b>           | <b>100 %</b> |
| Not automated (§6)       | 5            | 0                   | 0 %          |
| <b>Declared total</b>    | <b>52</b>    | <b>47</b>           | <b>90 %</b>  |

Coverage is computed over the requirements **declared in this document** and must not be confused with code coverage: it measures how many requirements have a bench verifying them, not how many lines are executed during the tests.

**Why two totals.** Reporting only the 100 % of automated requirements would be true and misleading at once: it is 100 % of what was *chosen* to be automated. The number that matters to an assessor is the second one — **47 requirements verified out of 52 declared** — and the five that are missing are listed by name in §6, not summarised into a percentage.

This table is checked by `tools/verifica-documenti.js`, which counts the identifiers actually present in the document and fails if the totals written here do not match. It is a check born of a real error: the totals said 36 over 9 scientific categories while the identifiers defined were 10, and `SCI-10` appeared twice — once as a verified requirement and once as an unverified one.

## 8. Declared coverage gaps

These are not defects: they are things that have **not been measured**, and that therefore cannot be asserted.

| Gap                                  | What it entails                                                                                                                                            | Why it is so                                                                                                                                                                                                                                                         |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Partial code coverage</b>         | Measured: <b>49.79 %</b> of statements over the widest path a bench walks. It is not the coverage of the whole battery, and it is statements, not branches | Chromium's profiler collects it inside the engine, with no build and without rewriting the source: the obstacle belonged to the tool ( <code>c8</code> , <code>istanbul</code> ), not to the problem. Value recorded; <code>audit_copertura</code> fails if it falls |
| <b>Chromium only</b>                 | Behaviour on Firefox and WebKit is verified by hand, not by a bench                                                                                        | The battery uses <code>playwright-core</code> , which downloads a single engine. In the verification environment the CDN for the other engines answers <b>403</b> to the network policy: they cannot be installed there                                              |
| <b>No visual regression</b>          | An unintended graphical change would not be caught                                                                                                         | Spectra are deterministic and comparable byte for byte ( <code>SCI-05</code> ): the comparison exists on the traces, not on the whole page                                                                                                                           |
| <b>2 entries without a structure</b> | Two entries out of 178 have no structure to show: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. It was six       | See <code>06-Scientific-Accuracy-Data-Provenance.md</code> §2.4                                                                                                                                                                                                      |
| <b>Performance on slow devices</b>   | First-paint time is not measured on low-end hardware                                                                                                       | It requires physical devices; the test is manual                                                                                                                                                                                                                     |

\_Document updated to version `bsi-v175`.\_